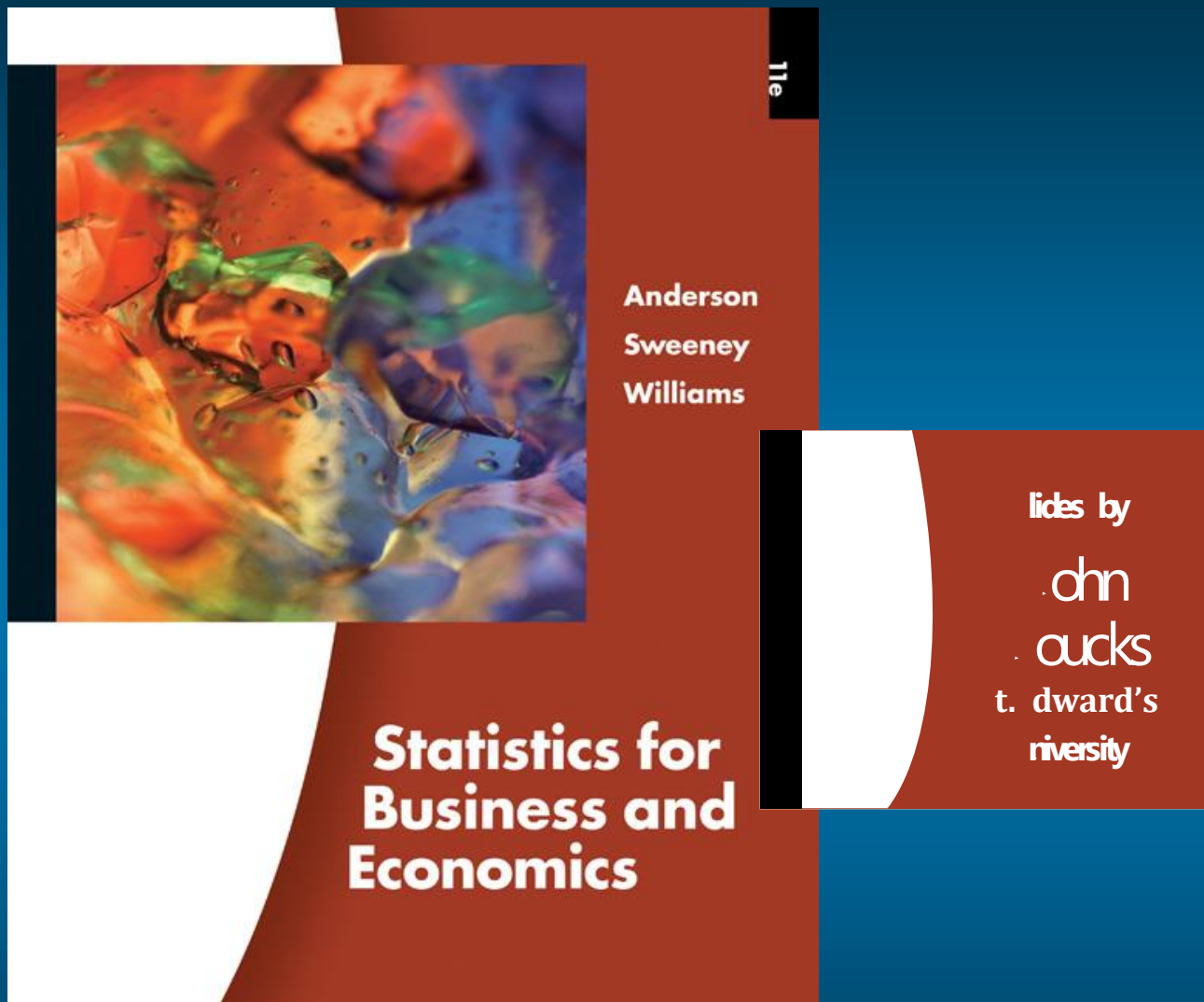




Chapter 18, Part A

Forecasting

- ▶ ■ Quantitative Approaches to Forecasting
- ▶ ■ Time Series Patterns
- ▶ ■ Forecast Accuracy
- ▶ ■ Moving Averages and Exponential Smoothing

Forecasting Methods

- ▶ ■ Forecasting methods can be classified as qualitative or quantitative.
- ▶ ■ Qualitative methods generally involve the use of expert judgment to develop forecasts.
- ▶ ■ Such methods are appropriate when historical data on the variable being forecast are either not applicable or unavailable.
- ▶ ■ We will focus exclusively on quantitative forecasting methods in this chapter.

Forecasting Methods

- ▶ ■ Quantitative forecasting methods can be used when:
 - past information about the variable being forecast is available,
 - the information can be quantified, and
 - it is reasonable to assume that the pattern of the past will continue into the future.
- ▶ ■ In such cases, a forecast can be developed using a time series method or a causal method.

Quantitative Forecasting Methods

- ▶ ■ Quantitative methods are based on an analysis of historical data concerning one or more time series.
- ▶ ■ A time series is a set of observations measured at successive points in time or over successive periods of time.
- ▶ ■ If the historical data used are restricted to past values of the series that we are trying to forecast, the procedure is called a time series method.
- ▶ ■ If the historical data used involve other time series that are believed to be related to the time series that we are trying to forecast, the procedure is called a causal method.

Time Series Methods

- ▶ ■ The objective of time series analysis is to discover a pattern in the historical data or time series and then extrapolate the pattern into the future.
- ▶ ■ The forecast is based solely on past values of the variable and/or past forecast errors.

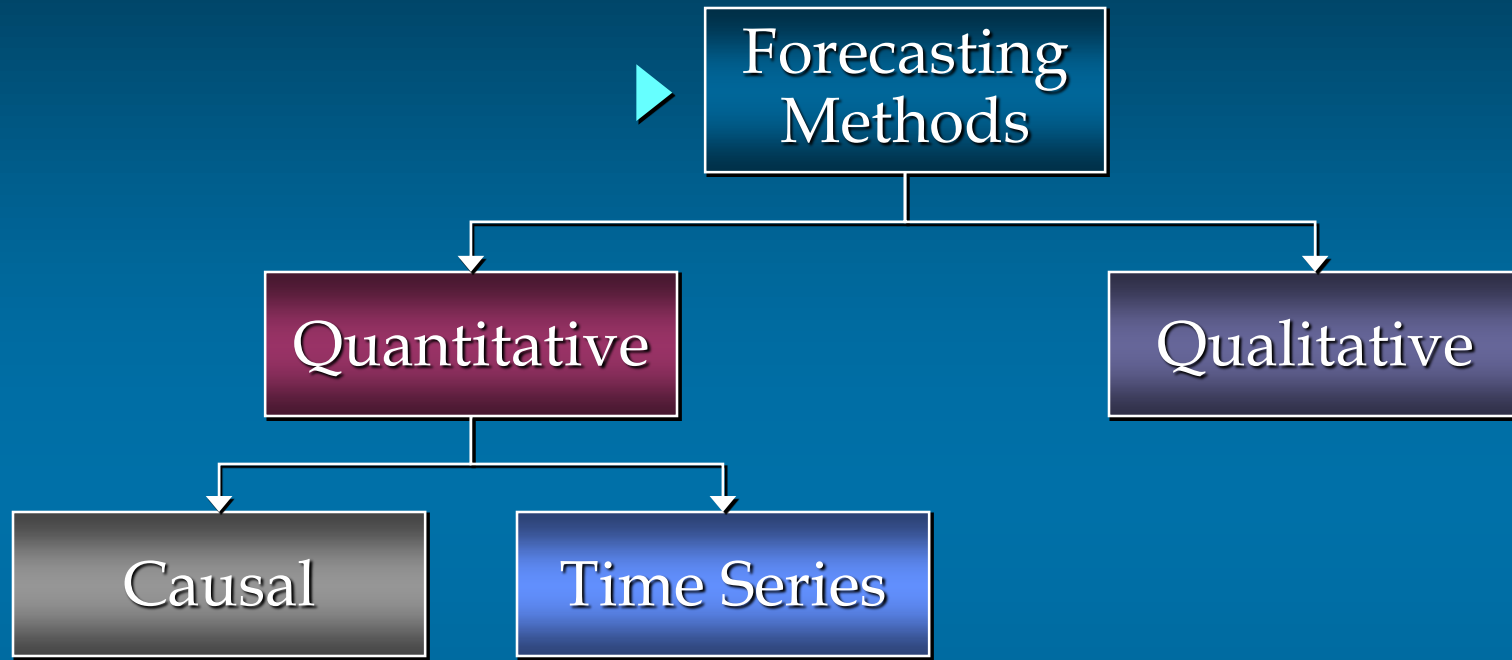
Causal Methods

- ▶ ■ Causal forecasting methods are based on the assumption that the variable we are forecasting has a cause-effect relationship with one or more other variables.
- ▶ ■ Looking at regression analysis as a forecasting tool, we can view the time series value that we want to forecast as the dependent variable.
- ▶ ■ If we can identify a good set of related independent, or explanatory, variables we may be able to develop an estimated regression equation for forecasting the time series.

Regression Analysis

- ▶ ■ By treating time as the independent variable and the time series as a dependent variable, regression analysis can also be used as a time series method.
- ▶ ■ Time-series regression refers to the use of regression analysis when the sole independent variable is time.
- ▶ ■ Cross-sectional regression refers to the use of regression analysis when the independent variable(s) is(are) something other than time.

Forecasting Methods



Time Series Patterns

- ▶ ■ A time series is a sequence of measurements taken every hour, day, week, month, quarter, year, or at any other regular time interval.
- ▶ ■ The pattern of the data is an important factor in understanding how the time series has behaved in the past.
- ▶ ■ If such behavior can be expected to continue in the future, we can use it to guide us in selecting an appropriate forecasting method.

Time Series Plot

- ▶ ■ A useful first step in selecting an appropriate forecasting method is to construct a time series plot.
- ▶ ■ A time series plot is a graphical presentation of the relationship between time and the time series variable.
- ▶ ■ Time is on the horizontal axis, and the time series values are shown on the vertical axis.

Time Series Patterns

- ▶ ■ The common types of data patterns that can be identified when examining a time series plot include:

Horizontal

Trend

Seasonal

Trend & Seasonal

Cyclical

Time Series Patterns

■ Horizontal Pattern

- ▶ • A horizontal pattern exists when the data fluctuate around a constant mean.
- ▶ • Changes in business conditions can often result in a time series that has a horizontal pattern shifting to a new level.
- ▶ • A change in the level of the time series makes it more difficult to choose an appropriate forecasting method.

Time Series Patterns

■ Trend Pattern

- ▶ • A time series may show gradual shifts or movements to relatively higher or lower values over a longer period of time.
- ▶ • Trend is usually the result of long-term factors such as changes in the population, demographics, technology, or consumer preferences.
- ▶ • A systematic increase or decrease might be linear or nonlinear.
- ▶ • A trend pattern can be identified by analyzing multiyear movements in historical data.

Time Series Patterns

■ Seasonal Pattern

- ▶ • Seasonal patterns are recognized by seeing the same repeating pattern of highs and lows over successive periods of time within a year.
- ▶ • A seasonal pattern might occur within a day, week, month, quarter, year, or some other interval no greater than a year.
- ▶ • A seasonal pattern does not necessarily refer to the four seasons of the year (spring, summer, fall, and winter).

Time Series Patterns

■ Trend and Seasonal Pattern

- ▶ • Some time series include a combination of a trend and seasonal pattern.
- ▶ • In such cases we need to use a forecasting method that has the capability to deal with both trend and seasonality.
- ▶ • Time series decomposition can be used to separate or decompose a time series into trend and seasonal components.

Time Series Patterns

■ Cyclical Pattern

- ▶ • A cyclical pattern exists if the time series plot shows an alternating sequence of points below and above the trend line lasting more than one year.
- ▶ • Often, the cyclical component of a time series is due to multiyear business cycles.
- ▶ • Business cycles are extremely difficult, if not impossible, to forecast.
- ▶ • In this chapter we do not deal with cyclical effects that may be present in the time series.

Selecting a Forecasting Method

- ▶ ■ The underlying pattern in the time series is an important factor in selecting a forecasting method.
- ▶ ■ Thus, a time series plot should be one of the first things developed when trying to determine what forecasting method to use.
- ▶ ■ If we see a horizontal pattern, then we need to select a method appropriate for this type of pattern.
- ▶ ■ If we observe a trend in the data, then we need to use a method that has the capability to handle trend effectively.

Forecast Accuracy

- ▶ ■ Measures of forecast accuracy are used to determine how well a particular forecasting method is able to reproduce the time series data that are already available.
- ▶ ■ Measures of forecast accuracy are important factors in comparing different forecasting methods.
- ▶ ■ By selecting the method that has the best accuracy for the data already known, we hope to increase the likelihood that we will obtain better forecasts for future time periods.

Forecast Accuracy

- ▶ ■ The key concept associated with measuring forecast accuracy is forecast error.

$$\text{Forecast Error} = \text{Actual Value} - \text{Forecast}$$

- ▶ ■ A positive forecast error indicates the forecasting method underestimated the actual value.
- ▶ ■ A negative forecast error indicates the forecasting method overestimated the actual value.

Forecast Accuracy

► ■ Mean Error

A simple measure of forecast accuracy is the mean or average of the forecast errors. Because positive and negative forecast errors tend to offset one another, the mean error is likely to be small. Thus, the mean error is not a very useful measure.

► ■ Mean Absolute Error (MAE)

This measure avoids the problem of positive and negative errors offsetting one another. It is the mean of the *absolute values* of the forecast errors.

Forecast Accuracy

► ■ Mean Squared Error (MSE)

This is another measure that avoids the problem of positive and negative errors offsetting one another. It is the average of the *squared* forecast errors.

► ■ Mean Absolute Percentage Error (MAPE)

The size of MAE and MSE depend upon the scale of the data, so it is difficult to make comparisons for different time intervals. To make such comparisons we need to work with relative or percentage error measures. The MAPE is the average of the *absolute percentage* errors of the forecasts.

Forecast Accuracy

- ▶ ■ To demonstrate the computation of these measures of forecast accuracy we will introduce the simplest of forecasting methods.
- ▶ ■ The naïve forecasting method uses the most recent observation in the time series as the forecast for the next time period.

$$F_{t+1} = \text{Actual Value in Period } t$$

Forecast Accuracy

■ Example: Rosco Drugs

► Sales of Comfort brand headache medicine for the past 10 weeks at Rosco Drugs are shown below.

► If Rosco uses the naïve forecast method to forecast sales for weeks 2 – 10, what are the resulting MAE, MSE, and MAPE values?

<u>Week</u>	<u>Sales</u>	<u>Week</u>	<u>Sales</u>
1	110	6	120
2	115	7	130
3	125	8	115
4	120	9	110
5	125	10	130

Forecast Accuracy

<u>Week</u>	<u>Sales</u>	<u>Naïve Forecast</u>	<u>Forecast Error</u>	<u>Absolute Error</u>	<u>Squared Error</u>	<u>Abs.% Error</u>
1	110					
2	115	110	5	5	25	4.35
3	125	115	10	10	100	8.00
4	120	125	-5	5	25	4.17
5	125	120	5	5	25	4.00
6	120	125	-5	5	25	4.17
7	130	120	10	10	100	7.69
8	115	130	-15	15	125	13.04
9	110	115	-5	5	25	4.55
10	130	110	20	20	400	15.38
			<u>Total</u>	80	850	65.35

Moving Averages

■ Naive Forecast Accuracy

▶ $MAE = \frac{80}{9} = 8.89$

▶ $MSE = \frac{850}{9} = 94.44$

▶ $MAPE = \frac{65.35}{9} = 7.26\%$

Moving Averages and Exponential Smoothing

- ▶ ■ Now we discuss three forecasting methods that are appropriate for a time series with a horizontal pattern:

Moving Averages

Weighted Moving Averages

Exponential Smoothing

- ▶ ■ They are called smoothing methods because their objective is to smooth out the random fluctuations in the time series.
- ▶ ■ They are most appropriate for short-range forecasts.

Moving Averages

- ▶ The moving averages method uses the average of the most recent k data values in the time series. As the forecast for the next period.

- ▶
$$F_{t+1} = \frac{\sum (\text{most recent } k \text{ data values})}{k} = \frac{Y_t + Y_{t-1} + \dots + Y_{t-k+1}}{k}$$

where:

F_{t+1} = forecast of the time series for period $t + 1$

- ▶ Each observation in the moving average calculation receives the same weight.

Moving Averages

- ▶ ■ The term *moving* is used because every time a new observation becomes available for the time series, it replaces the oldest observation in the equation.
- ▶ ■ As a result, the average will change, or move, as new observations become available.

Moving Averages

- ▶ ■ To use moving averages to forecast, we must first select the order k , or number of time series values, to be included in the moving average.
- ▶ ■ A smaller value of k will track shifts in a time series more quickly than a larger value of k .
- ▶ ■ If more past observations are considered relevant, then a larger value of k is better.

Moving Averages

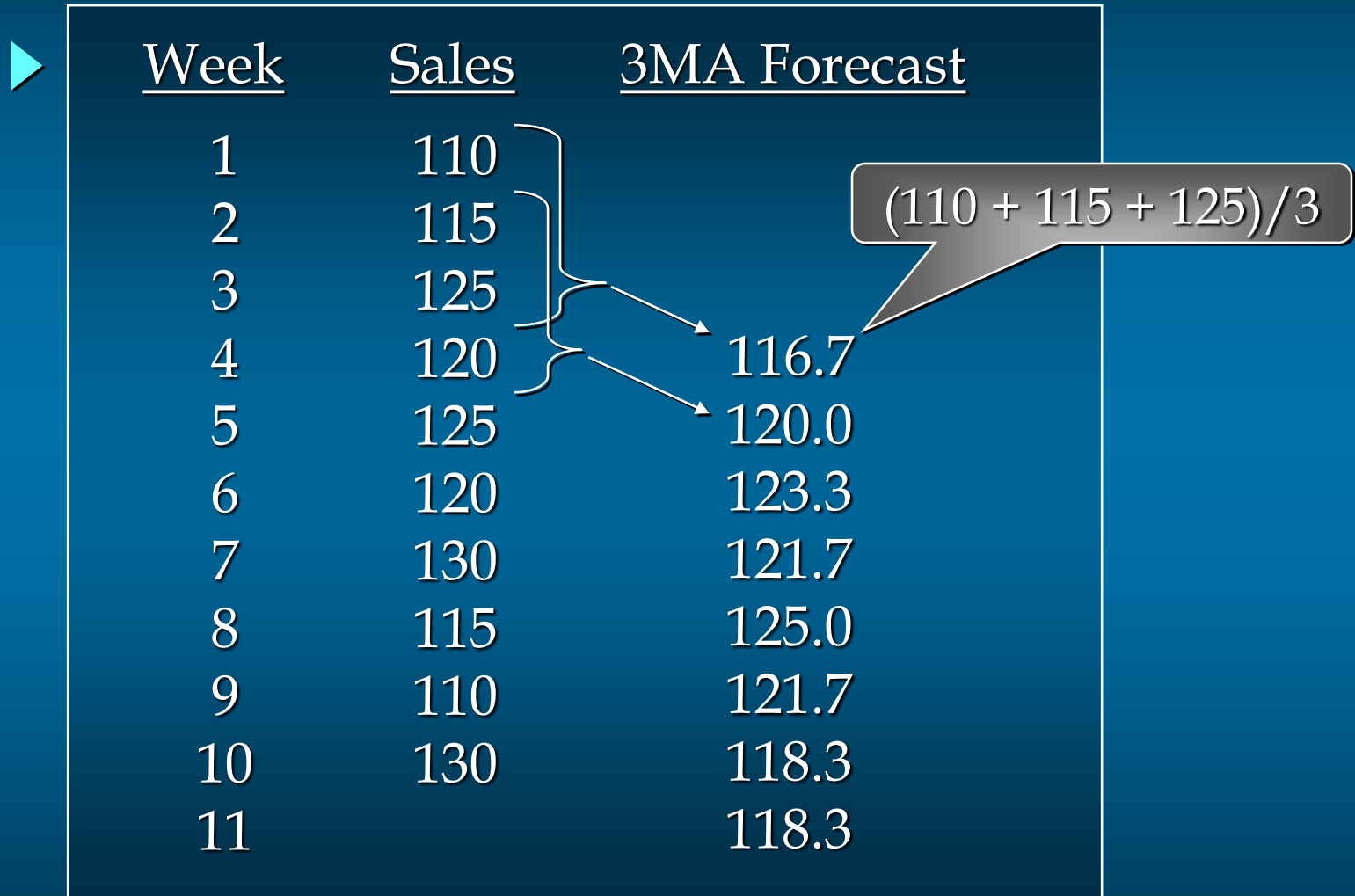
■ Example: Rosco Drugs

- ▶ If Rosco Drugs uses a 3-period moving average to forecast sales, what are the forecasts for weeks 4-11?

▶

<u>Week</u>	<u>Sales</u>	<u>Week</u>	<u>Sales</u>
1	110	6	120
2	115	7	130
3	125	8	115
4	120	9	110
5	125	10	130

Moving Averages



►

<u>Week</u>	<u>Sales</u>	<u>3MA Forecast</u>
1	110	
2	115	
3	125	
4	120	116.7
5	125	120.0
6	120	123.3
7	130	121.7
8	115	125.0
9	110	121.7
10	130	118.3
11		118.3

$(110 + 115 + 125)/3$

Moving Averages



<u>Week</u>	<u>Sales</u>	<u>3MA Forecast</u>	<u>Forecast Error</u>	<u>Absolute Error</u>	<u>Squared Error</u>	<u>Abs.% Error</u>
1	110					
2	115					
3	125					
4	120	116.7	3.3	3.3	10.89	2.75
5	125	120.0	5.0	5.0	25.00	4.00
6	120	123.3	-3.3	3.3	10.89	2.75
7	130	121.7	8.3	8.3	68.89	6.38
8	115	125.0	-10.0	10.0	100.00	8.70
9	110	121.7	-11.7	11.7	136.89	10.64
10	130	118.3	11.7	11.7	136.89	9.00
		<u>Total</u>	6.6	53.3	489.45	44.22

Moving Averages

■ 3-MA Forecast Accuracy

$$\blacktriangleright \text{MAE} = \frac{53.3}{7} = 7.61$$

$$\blacktriangleright \text{MSE} = \frac{489.45}{7} = 69.92$$

$$\blacktriangleright \text{MAPE} = \frac{44.22}{7} = 6.32\%$$

The 3-week moving average approach provided more accurate forecasts than the naïve approach.

Weighted Moving Averages

■ Weighted Moving Averages

- ▶ • To use this method we must first select the number of data values to be included in the average.
- ▶ • Next, we must choose the weight for each of the data values.
 - The more recent observations are typically given more weight than older observations.
 - For convenience, the weights should sum to 1.

Weighted Moving Averages

■ Weighted Moving Averages

- ▶ • An example of a 3-period weighted moving average (3WMA) is:

$$3WMA = .2(110) + .3(115) + .5(125) = 119$$

Weights (.2, .3, and .5) sum to 1

Most recent of the three observations

Exponential Smoothing

- ▶ • This method is a special case of a weighted moving averages method; we select only the weight for the most recent observation.
- ▶ • The weights for the other data values are computed automatically and become smaller as the observations grow older.
- ▶ • The exponential smoothing forecast is a weighted average of all the observations in the time series.
- ▶ • The term *exponential smoothing* comes from the exponential nature of the weighting scheme for the historical values.

Exponential Smoothing

■ Exponential Smoothing Forecast

►
$$F_{t+1} = \alpha Y_t + (1 - \alpha)F_t$$

where:

F_{t+1} = forecast of the time series for period $t + 1$

Y_t = actual value of the time series in period t

F_t = forecast of the time series for period t

α = smoothing constant ($0 \leq \alpha \leq 1$)

and let:

$F_2 = Y_1$ (to initiate the computations)

Exponential Smoothing

■ Exponential Smoothing Forecast

- ▶ • With some algebraic manipulation, we can rewrite $F_{t+1} = \alpha Y_t + (1 - \alpha)F_t$ as:

$$F_{t+1} = F_t + \alpha(Y_t - F_t)$$

- ▶ • We see that the new forecast F_{t+1} is equal to the previous forecast F_t plus an adjustment, which is α times the most recent forecast error, $Y_t - F_t$.

Exponential Smoothing

■ Example: Rosco Drugs

- If Rosco Drugs uses exponential smoothing to forecast sales, which value for the smoothing constant α , .1 or .8, gives better forecasts?

►

<u>Week</u>	<u>Sales</u>	<u>Week</u>	<u>Sales</u>
1	110	6	120
2	115	7	130
3	125	8	115
4	120	9	110
5	125	10	130

Exponential Smoothing

■ Using Smoothing Constant Value $\alpha = .1$



$$\begin{aligned} F_2 &= Y_1 &= 110 \\ F_3 &= .1Y_2 + .9F_2 = .1(115) + .9(110) &= 110.5 \\ F_4 &= .1Y_3 + .9F_3 = .1(125) + .9(110.5) &= 111.95 \\ F_5 &= .1Y_4 + .9F_4 = .1(120) + .9(111.95) &= 112.76 \\ F_6 &= .1Y_5 + .9F_5 = .1(125) + .9(112.76) &= 113.98 \\ F_7 &= .1Y_6 + .9F_6 = .1(120) + .9(113.98) &= 114.58 \\ F_8 &= .1Y_7 + .9F_7 = .1(130) + .9(114.58) &= 116.12 \\ F_9 &= .1Y_8 + .9F_8 = .1(115) + .9(116.12) &= 116.01 \\ F_{10} &= .1Y_9 + .9F_9 = .1(110) + .9(116.01) &= 115.41 \end{aligned}$$

Exponential Smoothing

■ Using Smoothing Constant Value $\alpha = .8$



$$F_2 = \quad \quad \quad = 110$$

$$F_3 = .8(115) + .2(110) \quad = 114$$

$$F_4 = .8(125) + .2(114) \quad = 122.80$$

$$F_5 = .8(120) + .2(122.80) = 120.56$$

$$F_6 = .8(125) + .2(120.56) = 124.11$$

$$F_7 = .8(120) + .2(124.11) = 120.82$$

$$F_8 = .8(130) + .2(120.82) = 128.16$$

$$F_9 = .8(115) + .2(128.16) = 117.63$$

$$F_{10} = .8(110) + .2(117.63) = 111.53$$

Exponential Smoothing ($\alpha = .1$)



<u>Week</u>	<u>Sales</u>	$\alpha = .1$ <u>Forecast</u>	<u>Forecast</u> <u>Error</u>	<u>Absolute</u> <u>Error</u>	<u>Squared</u> <u>Error</u>	<u>Abs.%</u> <u>Error</u>
1	110					
2	115	110.00	5.00	5.00	25.00	4.35
3	125	110.50	14.50	14.50	210.25	11.60
4	120	111.95	8.05	8.05	64.80	6.71
5	125	112.76	12.24	12.24	149.94	9.79
6	120	113.98	6.02	6.02	36.25	5.02
7	130	114.58	15.42	15.42	237.73	11.86
8	115	116.12	-1.12	1.12	1.26	0.97
9	110	116.01	-6.01	6.01	36.12	5.46
10	130	115.41	14.59	14.59	212.87	11.22
			<u>Total</u>	82.95	974.22	66.98

Exponential Smoothing ($\alpha = .1$)

■ Forecast Accuracy

$$\blacktriangleright \text{MAE} = \frac{82.95}{9} = 9.22$$

$$\blacktriangleright \text{MSE} = \frac{974.22}{9} = 108.25$$

$$\blacktriangleright \text{MAPE} = \frac{66.98}{9} = 7.44\%$$

Exponential smoothing (with $\alpha = .1$) provided less accurate forecasts than the 3-MA approach.

Exponential Smoothing ($\alpha = .8$)



<u>Week</u>	<u>Sales</u>	$\alpha = .8$ <u>Forecast</u>	<u>Forecast</u> <u>Error</u>	<u>Absolute</u> <u>Error</u>	<u>Squared</u> <u>Error</u>	<u>Abs.%</u> <u>Error</u>
1	110					
2	115	110.00	5.00	5.00	25.00	4.35
3	125	114.00	11.00	11.00	121.00	8.80
4	120	122.80	-2.20	2.20	7.84	1.83
5	125	120.56	4.44	4.44	19.71	3.55
6	120	124.11	-4.11	4.11	16.91	3.43
7	130	120.82	9.18	9.18	84.23	7.06
8	115	128.16	-13.16	13.16	173.30	11.44
9	110	117.63	-7.63	7.63	58.26	6.94
10	130	111.53	18.47	18.47	341.27	14.21
			<u>Total</u>	75.19	847.52	61.61

Exponential Smoothing ($\alpha = .8$)

■ Forecast Accuracy

$$\blacktriangleright \text{MAE} = \frac{75.19}{9} = 8.35$$

$$\blacktriangleright \text{MSE} = \frac{847.52}{9} = 94.17$$

$$\blacktriangleright \text{MAPE} = \frac{61.61}{9} = 6.85\%$$

Exponential smoothing (with $\alpha = .8$) provided more accurate forecasts than ES with $\alpha = .1$, but less accurate than the moving average (with $k = 3$).

End of Chapter 18, Part A

